JONATHAN M. CARCACHE

Davie West
3200 College Ave
Davie, FL 33314

Florida Atlantic University
Integrative Biology

Jcarcache2025@fau.edu
Jcarcache08@outlook.com
Website: https://jcarcache.github.io/

Education

Ph.D. **Integrative Biology** 2025 –
Florida Atlantic University, Davie, FL
Committee: **Jacob Francis**, Daniel Wieczynski, Luis Y. Rosario-Santiago

B.S. **Biology** 2022
Cum Laude
Florida International University, Miami, FL
Concentration in Plant Sciences
Advisor: **Adam Roddy**

A.A. **Environmental Science** 2018
Miami Dade College, Miami, FL

Fellowships and Awards

2026 – NSF, National Science Foundation Graduate Research Fellowship - - - - - - - - - - - - **HM**
2026 – OBFS, Maxwell/Hanrahan Field Biology Research Grant - - - - - - - - - - - - - - - - - **$5,000**
2025 – FAU, Vincent Saurino Fellowship for Grad Students in Biological Science - - - - - $**2,537**
2022 – FIU**,** Hannah Kelly Foundation Tropical Botany Scholarship - - - - - - - - - - - - - - - - - **$500**
2021 – FIU, Tropical Fern and Exotic Plant Society Scholarship - - - - - - - - - - - - - - - - - - - **$500**
2021 – BSA, Botanical Society of America PLANTS Grants - - - - - - - - - - - - - - - - - **Recipient**

Research Appointments

Lead "Mixed" Technician 2025
National Ecological Observatory Network, D18 & D19, Fairbanks/Toolik, AK.

NPS SFCN Inventory & Monitoring Intern 2022 - 2024
Terrestrial Sampling Team, Miami, FL.

Lead Field Technician 2023
Arctic Observation Network & International Tundra Experiment, Plant Physiology Oberbaurer lab, Toolik Field station, AK.

Undergraduate Honors Thesis 2021 - 2022
Florida International University, Department of Biological Sciences, Roddy Plant Physiology Lab, Miami, FL.

Research Intern 2021
Harvard Forest, Frey Lab, Petersham, MA

Fairchild Tropical Botanic Garden (FTBG) Intern 2021
FTBG, Plant Conservation team, Miami, FL

Lemur Conservation Intern 2020
Tropical Conservation Institute: International Research Experience for Students, Miami, FL

Field Botanist 2019-2020
FIU Tree Survey, Franck Plant Taxonomy Lab, Miami, FL

Student Lab Lead 2017-2018
Miami Dade College Orchid Propagation Lab, Miami, FL

Publications

Published

1. Williams, J.N., Barker, D.A., Millet, A.J., **Carcache, J.M.** and Francis, J.S. (2026), Petal infrared transmission warms flowers and reduces microbial abundance. New Phytol. https://doi.org/10.1111/nph.71608

2. Francis, Jacob S., **Jonathan M. Carcache**, and Naomi L. Sananikone. "A subtropical range extension of the entomopathogenic fungus *Eryniopsis lampyridarum* infecting *Chauliognathus* spp." *Journal of Natural History* 60.21-24 (2026): 1193-1196.

Mentorships

Adam Bloom- Undergraduate Field Technician	Florida Atlantic University; Toolik Field Station, Alaska.	2026
Emily Panton- Undergraduate Field Technician	Florida Atlantic University; Toolik Field Station, Alaska.	2026
Kyla Morris- Undergraduate Field Technician	Marietta College; Toolik Field Station, Alaska	2023

Poster Presentations

Carcache, J.C., Rosario-Santiago, L.Y., Francis, J.S (2026, Aug.02-05). Good “parasites”: How can context shift the balance of plant-plant symbiosis? Botanical Society of America poster session, Tucson, AZ, USA.

Carcache, J.C., Romero-Olivares, A., Serita, F. (2021, August 5th). Fungi of the Future:

Investigating Dominant Fungal Taxa Throughout Global Change Treatments. Harvard Forest 28th Annual Student Symposium

Media and Outreach

- 2026- Miami Herald: This Arctic plant has a bad rep. Florida scientists are trying to exonerate it
- 2026 - AP News: A lush knee-high jungle grows in the Arctic, but warming is changing Earth's mini-greenhouse
- 2026 - Geophysical Institute: Seeking the benefits of a parasite
- 2026- Toolik Newsletter- Toolik Field Station Spring 2026 Newsletter